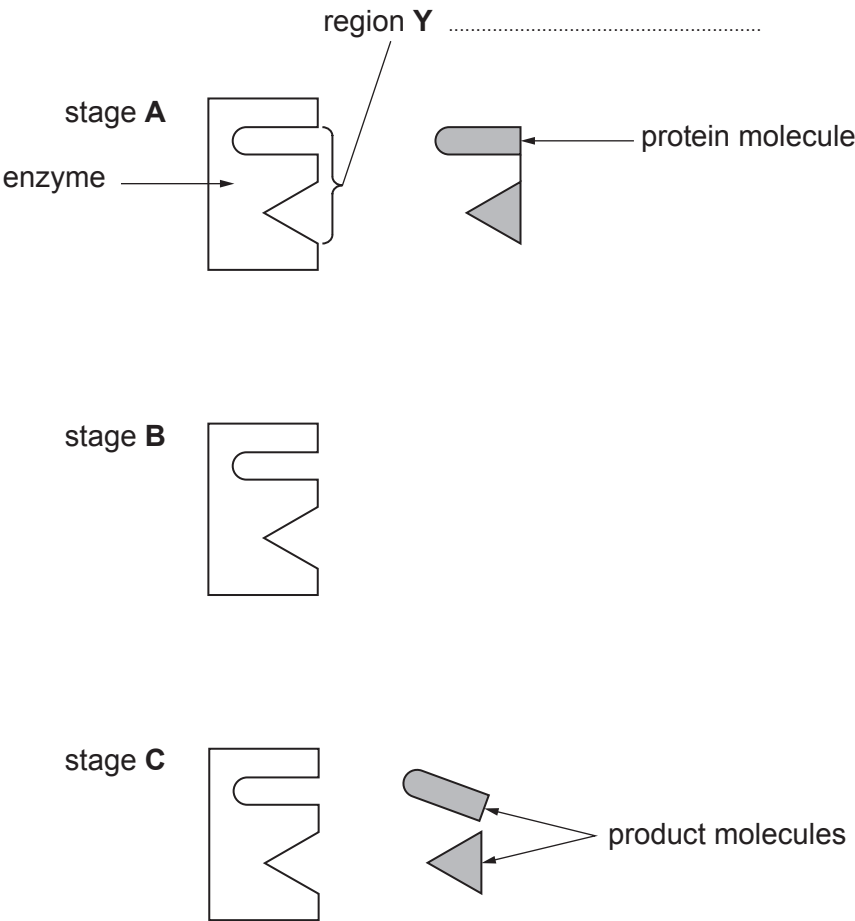


2. The lock and key model is used to explain how enzymes work. It was first suggested in 1894 by Emile Fischer.

The diagrams below show stages in the action of an enzyme which digests protein molecules.



- (a) (i) **Label region Y** on the diagram. [1]
- (ii) **Complete the diagram** to show what occurs at stage B. [1]



Examiner
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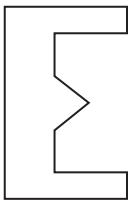
(b) (i) State the name of the **type** of enzyme shown in stage **A**. [1]

.....

(ii) State the name of the product molecules shown in stage **C**. [1]

.....

(c) The diagram below shows the same enzyme but it has been denatured.



(i) Explain why this denatured enzyme would not function. [2]

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(ii) Suggest how the enzyme molecule shown could have become denatured. [1]

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